

National University of Computer & Emerging Sciences

Homework # 1e

Measure of Central Tendency

1. The mean weight of 25 children in a class is 58 kg. The mean weight of a second class of 29 children is 62 kg. Find the mean weight of all the children.
2. In marketing, why might a company focus on the mode of customer preferences rather than the mean or median?
3. The commute times (in minutes) for employees of a company in Pakistan are summarized in the table below:

Commute Time	Number of Employees
Less than 10	14
10 but less than 20	26
20 but less than 30	64
30 but less than 40	46
40 but less than 50	28
50 but less than 60	16
60 but less than 70	8
70 but less than 80	4

- i. Calculate the **mean**, **median**, and **mode** of the employees' commute times based on the provided data. Ensure that the calculations account for the grouped nature of the data.
 - ii. The company is considering negotiations regarding compensating employees for their commute time to work. If you were:
 - **The employer**, explain which measure of central tendency you would use to support your case and justify your choice.
 - **A trade union representative**, explain which measure of central tendency you would use to argue for the employees and justify your choice.
4. If a and b are positive numbers show that

$$HM = \frac{GM^2}{AM}$$

Additionally, find the AM , GM and HM for numbers 45, 21, 77, and 68.